**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# LESSON 278 · SQL SELECT, JOIN and nested SELECT

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

## Do Now · Recall previous learning

Q1. What is the purpose of an INNER JOIN?

Q2. Which relational key connects two tables?

Q3. What does WHERE do?

Q4. Binary search for 9 in sorted [1,3,5,7,9,11,13]. Show midpoint indexes.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# I Do · Read new knowledge

## Key vocabulary

SELECT — Choose fields for output.

INNER JOIN — Combine matching rows.

nested SELECT — Query inside another query.

IN — Compare value against returned set.

WHERE — Filter rows.

## New learning

Understand — SELECT names output columns, FROM identifies input tables and WHERE restricts rows using a Boolean predicate.

Trace — INNER JOIN combines rows matching a join condition. A nested SELECT can return a set of keys used by an outer IN condition.

Explain — Both example queries return Ada and Ben, but joining is more flexible when fields from both tables are required.

## Worked example

Write SQL for student names where Score.Mark>80.

1. Predict — Only Ben qualifies because score 82 is the sole result above 80.

2. Check the answer — Explain which rule makes this result correct.

# We Do · Practise with guidance

Write a nested SELECT for student names scoring at least 75.

```
SELECT Name FROM Student WHERE ID IN
(SELECT ID FROM Score WHERE Mark>=75);
```

Predict

Subquery returns IDs 1 and 2

Check the answer

outer query returns Ada and Ben.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

### Question 1

Write SELECT to show all Name values in Student sorted by Name ascending. [4]

### Question 2

Write and trace a join returning Name and Mark for scores >=70. [6]

### Question 3

Write a nested SELECT for students scoring above 75. Explain result. [6]

# Exit Ticket · Answer before checking

INNER JOIN results can be the same as a nested key SELECT for a particular data set.

**BLACK PEN** Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

# Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

### Page 1 onwards · Do Now

Q1. Combines matching rows.

Q2. Foreign key referencing primary key.

Q3. Filters results according to a predicate.

Q4. low0 high6 mid3→7 then low4 high6 mid5→11 then high4 mid4→9, found index4.

### You Do pages · Independent questions

Q1. SELECT Name [1] FROM Student [1] ORDER BY Name (ascending) [2]; explicitly writing ASC is optional because SQL defaults to ascending order.

Q2. SELECT Student.Name,Score.Mark [1] FROM Student [1] INNER JOIN Score [1] ON Student.ID=Score.ID [1] WHERE Score.Mark>=70 [1]; output Ada75, Ben82 [1].

Q3. SELECT Name FROM Student WHERE ID IN (SELECT ID FROM Score WHERE Mark>75) [4]; inner query returns only ID2 [1], outer returns Ben [1].

### Exit Ticket · Model answer

True. Different query forms can yield equivalent selected results.

# Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

**1. Copy the whole original question:**

**2. Write a complete new answer in GREEN PEN. Show working if needed:**

**BLACK PEN** Answer all questions first.

**GREEN PEN** Then: tick correct answers; rewrite mistakes.

**Do Now** Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check my understanding